

Common PCB Design Mistakes

Exhaustive Study Notes for Hardware Engineering

1. The True Cost of a Design Error

In software, a bug can be patched with an over-the-air update in minutes. In hardware, a design mistake forces a board respin. This costs thousands of dollars in NRE (Non-Recurring Engineering) fees, wastes weeks of lead time, and can entirely derail a product launch. Understanding common traps is the first step to avoiding them.

2. The Consolidated List: Most Frequent Mistakes

A. Schematic & Footprint Errors

- **Incorrect Pin Mapping:** Relying blindly on community libraries where schematic pins do not accurately map to the physical footprint pads (e.g., swapping Source and Drain on a MOSFET).
- **Mismatched Mechanical Dimensions:** Designing a footprint based on the "typical" dimensions rather than accounting for maximum tolerances, resulting in components that literally do not fit on the board.
- **Ignoring Component Derating:** Specifying a 10V capacitor for a 9V rail. Ceramic capacitors (especially Class II like X7R) lose significant capacitance as DC bias approaches their rated voltage.

B. Layout & Routing Errors

- **Routing Over Split Planes:** Routing a high-speed digital trace over a gap in the reference ground plane. This destroys the return path, drastically increasing loop inductance and turning the trace into an EMI radiating antenna.
- **Decoupling Capacitors Too Far:** Placing bypass capacitors far from the IC power pins or tying them to ground with long, thin traces. This renders them useless for supplying high-frequency transient current.
- **Starving Pads (Vias in Pads):** Placing an untented via directly inside an SMT pad without having it filled and plated over. During reflow, the via acts as a capillary tube, sucking the solder away and causing a disconnected or "tombstoned" component.

C. Manufacturing & DFM Errors

- **Missing Thermal Reliefs:** Connecting through-hole components or small SMT pads directly to massive copper planes without thermal spokes. The plane acts as a heatsink during assembly, causing cold, unreliable solder joints.

- **Silkscreen Over Pads:** Allowing silkscreen text or lines to overlap with exposed copper pads. The ink prevents solder paste from adhering properly, ruining the electrical connection.
- **Acid Traps:** Creating acute (less than 90-degree) trace angles. During the etching process, acid can pool in these tight corners, over-etching the copper and creating an open circuit. (Note: Modern fabs handle this better, but it remains a poor practice).

3. Building Bulletproof Habits

Mistakes are prevented by process, not just knowledge. To transition from a beginner to a reliable professional, build these habits:

- **The 1:1 Paper Test:** Before sending a board to the fab, print the layout on standard paper at exactly 1:1 scale. Place your physical components on the paper to verify footprints, connector pitches, and mechanical clearances.
- **Trust No Datasheet Blindly:** Always verify library footprints against the mechanical section of the manufacturer's official datasheet. Never assume a downloaded library is flawless.
- **Run DRC Iteratively:** Do not wait until the layout is 100% finished to run the Design Rule Check (DRC). Run it frequently during the layout process to catch spacing and clearance violations before they snowball into massive rerouting efforts.
- **Formal Checklists:** Always use a standardized pre-release checklist (like the one covered in Day 37) to systematically eliminate "silent assumptions."